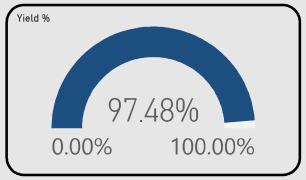
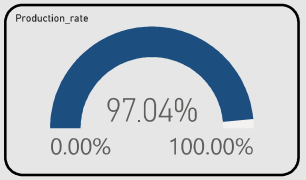
Production Performance Dashboard-2024

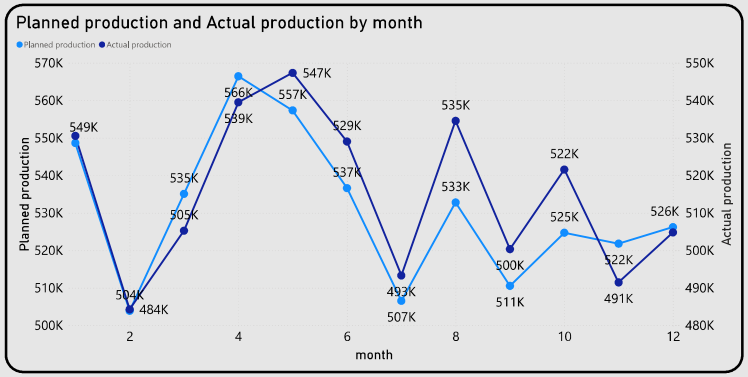
6M	195K	156K	3.16%	472K	2.52%
Total production	Total scrap	Rework units	scrap_rate	Total downtime	rework_rate

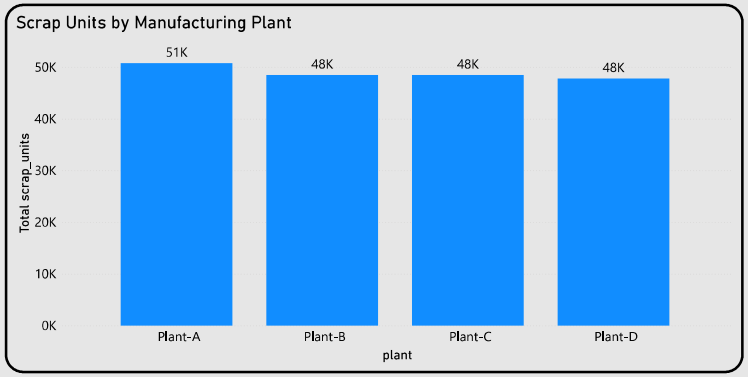


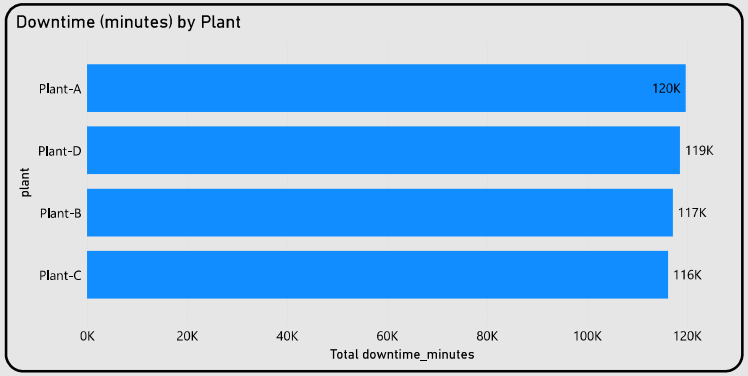


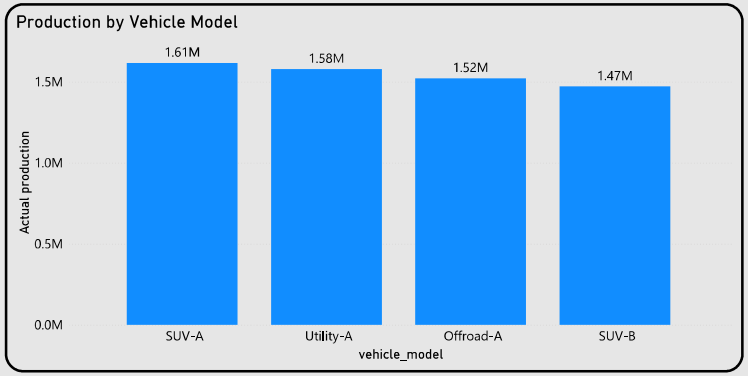
vehicle_model
- Offroad-A
- SUV-A
- SUV-B
- Utility-A

plant
- Plant-A
- Plant-B
- Plant-C
- Plant-D









1. **Plant-A recorded the highest downtime**, which could impact production efficiency and output stability.
2. **Scrap generation is highest in Plant-A**, indicating potential process quality issues in that facility.
3. **Production targets were closely met in the first half of the year**, while the gap between planned and actual production increased in the later months.
4. **Utility-A and SUV-A platforms contribute significantly to overall production volume.**
5. Improving **downtime management in Plant-C could improve overall production efficiency by 2–3%.**

Manufacturing Quality Analysis-2024

2.43% Defect rate

150K Total defects

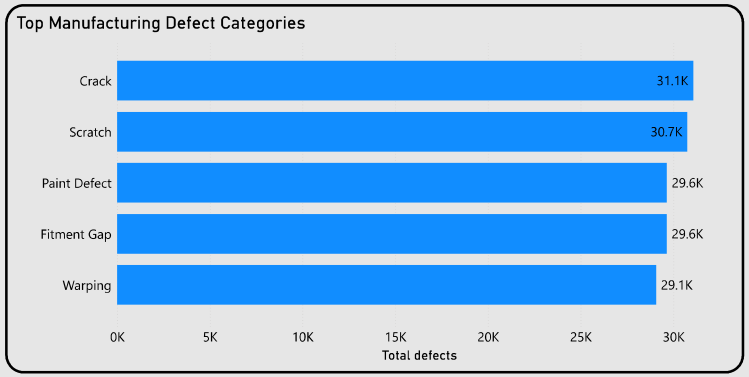
₹ 80.05M Rework cost

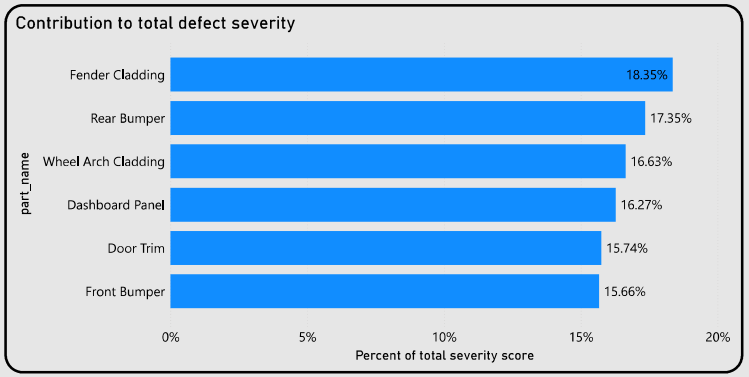
Plant
- Plant-A
- Plant-B
- Plant-C
- Plant-D

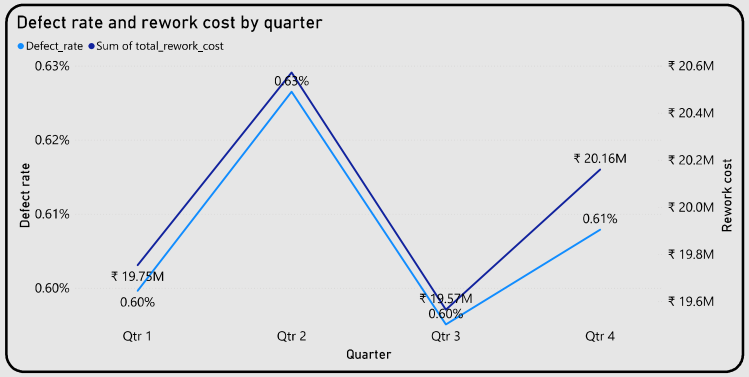
vehicle_model
- Offroad-A
- SUV-A
- SUV-B
- Utility-A

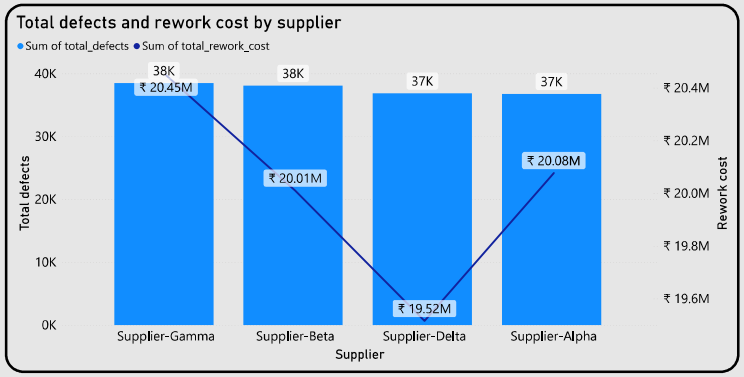
supplier
- Supplier-Alpha
- Supplier-Beta
- Supplier-Delta
- Supplier-Gamma

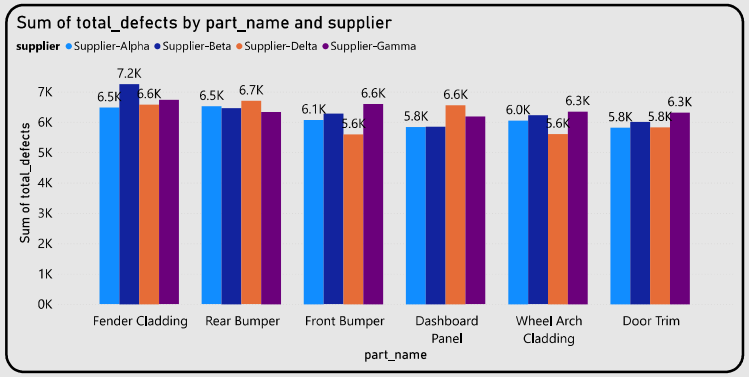
part_name
- Dashboard Panel
- Door Trim
- Fender Cladding
- Front Bumper
- Rear Bumper

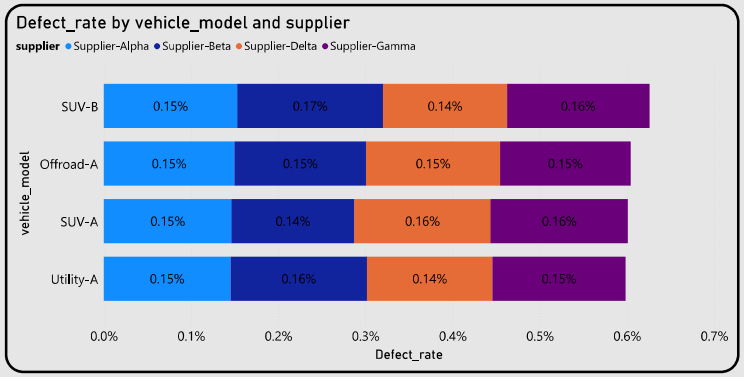












1. **Fender Cladding and Rear Bumper components show the highest defect counts**, indicating these parts require process quality improvements.
2. **Supplier-Gamma and Supplier-Beta contribute the largest share of defects**, suggesting supplier process audits may be required.
3. **Scratch and Paint Defects are the most common defect categories**, indicating surface finishing issues in production.
4. **Defect rates remain relatively stable across quarters**, but rework cost shows slight fluctuations due to severity of defects.
5. **Parts contributing highest severity scores should be prioritized for design and process improvements.**

Warranty Claims & Repair Cost Analysis

32K
Total claims

₹ 107.90M
Total repair cost

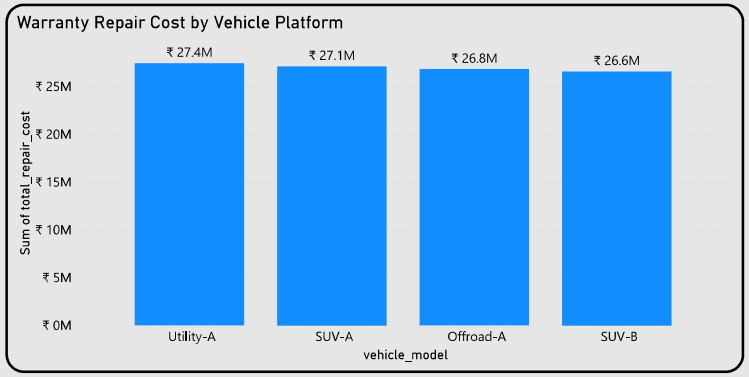
0.52%
Warranty_rate

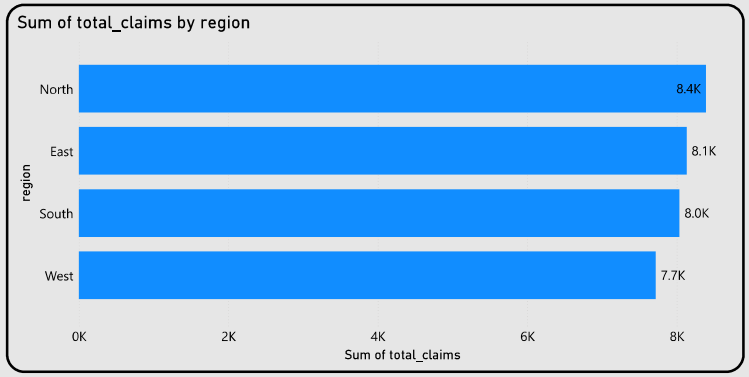
plant
- [ ] Plant-A
- [ ] Plant-B
- [ ] Plant-C
- [ ] Plant-D

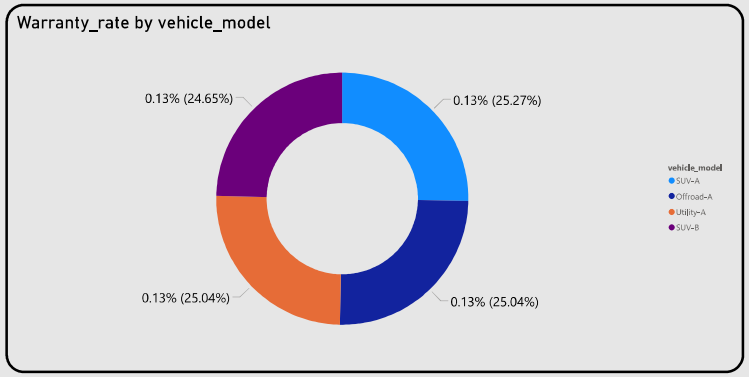
vehicle_model
- [ ] Offroad-A
- [ ] SUV-A
- [ ] SUV-B
- [ ] Utility-A

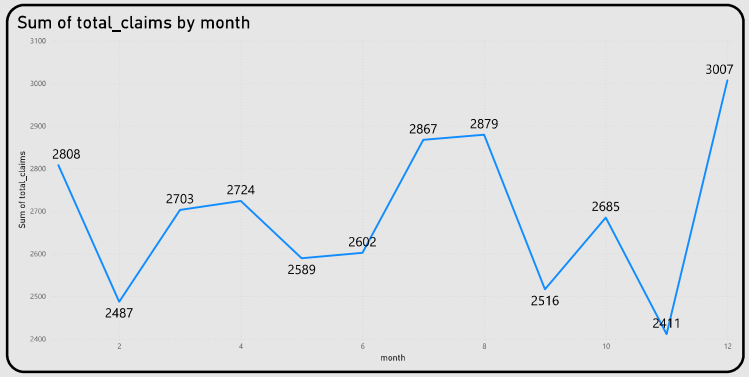
supplier
- [ ] Supplier-Alpha
- [ ] Supplier-Beta
- [ ] Supplier-Delta
- [ ] Supplier-Gamma

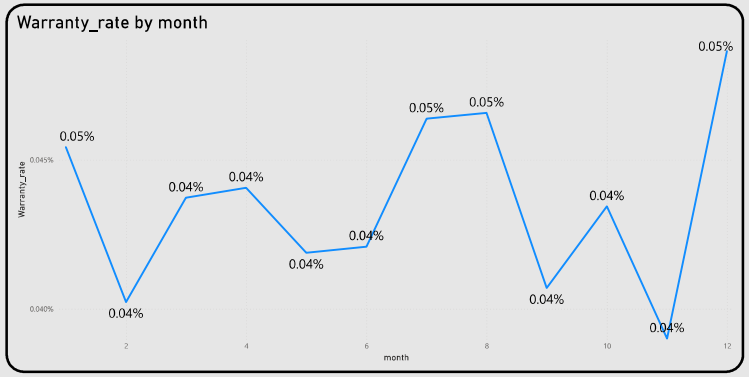
region
- [ ] East
- [ ] North
- [ ] South
- [ ] West

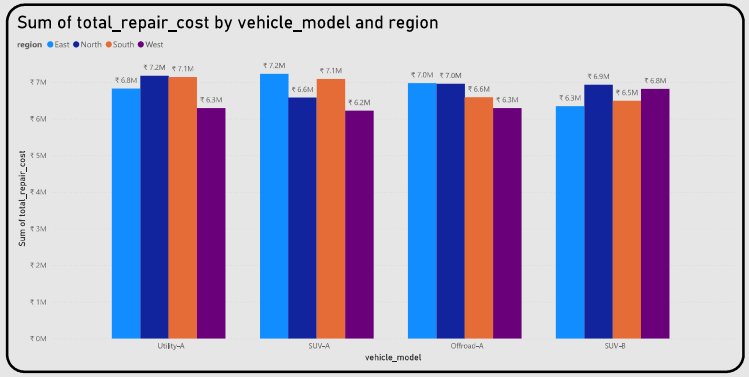












1. **Offroad-A platform generates the highest warranty repair cost**, suggesting reliability issues in specific components.
2. **North region reports the highest number of warranty claims**, possibly due to higher vehicle sales or harsher operating conditions.
3. **Warranty claims increase during July and December**, which may correspond to seasonal driving conditions or higher vehicle usage.
4. **SUV platforms show moderate claim frequency but lower repair cost compared to Offroad platforms.**
5. **Targeted improvements in high-cost components of Offroad-A could significantly reduce warranty expenditure.**